

Problem Statement No.7: Research Agent

The Challenge

Researchers, students, and professionals often deal with scattered academic papers, technical articles, and experimental data spread across multiple platforms. Filtering relevant knowledge, summarizing insights, and keeping track of the latest developments is time-consuming and inefficient. The challenge is not just to collect references but to **synthesize, analyze, and generate actionable research insights** in real time.

The Objective

Build an **agentic AI application** that acts as an intelligent research companion. The solution should:

1. **Fuse Academic Sources:** Aggregate data from research papers, journals, preprints, and knowledge repositories, then synthesize them into clear, concise literature reviews.
2. **Enable Multimodal Research Input:** Accept user queries, notes, charts, or even PDFs, and convert them into structured insights, references, and summaries.
3. **Create a Predictive & Agentic Layer:** Highlight emerging trends, detect citation gaps, and predict promising directions for future research in a given domain.
4. **Visualize Research Insights:** Provide an interactive dashboard with topic clusters, citation maps, sentiment/trend analysis, and concise knowledge graphs.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models